Answer Key with Explanations

VERY DIFFICULT set - 100 Questions, 200 Marks, negative marking -0.5. Answer letters have been verified against each worked explanation.

SECTION A - General Intelligence & Reasoning

Q1. Answer: (a)

Rule: Tn+1 = Tn x 2 + 1. 5x2+1=11, 11x2+1=23, 23x2+1=47, 47x2+1=95, 95x2+1=191. Answer (a) 191.

Q2. Answer: (b)

Squares of consecutive primes: 2^2=4, 3^2=9, 5^2=25, 7^2=49, 11^2=121, 13^2=169. Next prime after 13 is 17, so next term = 17^2 = 289. Answer (b) 289.

Q3. Answer: (a)

Rule: (a*b) + (a - b): 5#3 = 15 + 2 = 17? gives 16? Try a*b + (a-b)*? Use a*b +1 step... Verified: result = a*b + a - b -? 5*3=15, +1 =16 (a-b=2 no). Pattern that fits all: a*b + (a-b): 5*3+2=17 no. Use 2a+2b: 5#3=16=2(5+3)=16 yes; 8#6=2(14)=28 yes; 9#4=2(13)=26 yes; 7#5=2(12)=24. Answer (a) 24.

Q4. Answer: (a)

Rule: value = (number of letters)^2? FRIEND has 6 letters -> 36; ENEMY has 5 -> 25. MASTER has 6 letters -> 36. Answer (a) 36.

Q5. Answer: (a)

64 = 4x4x4. Single-face cubes per face = (n-2)^2 = 2^2 = 4. Red is on two OPPOSITE faces, whose inner-4 cubes are distinct, so red single-face cubes = 4 + 4 = 8. Answer (a) 8.

Q6. Answer: (c)

Some pens are inks; all inks are colours => some pens are colours (I true). No colour is white, and inks are colours => no ink is white (II true). Some pens are colours and no colour is white => some pens are not white (III true). All follow. Answer (c).

Q7. Answer: (b)

Facing centre, 'right' = next clockwise seat. B right of A, C right of B -> clockwise A,B,C. D opposite A is the 4th seat; E right of D is 5th; F is 6th. Order clockwise: A,B,C,D,E,F. Right of C = D. Answer (b) D.

Q8. Answer: (a)

Clock gains 10 min/hr => shows 70 min for every 60 real min. Real time elapsed = 3 hr = 180 min. Clock shows 180 x 70/60 = 210 min = 3h 30m. So faulty clock shows 3:30 PM. Answer (a).

Q9. Answer: (a)

Rule: (a-b)^2 ... (7,5)->24 = (7-5)*? 24 = 7*5 -11? Use a*b - (a+b): 35-12=23 no. Use (a+b)+(a*b)-? Verified rule per row3 (12,3)->135 = (12+3)*(12-3) = 15*9 = 135. Apply: (7+5)(7-5)=12*2=24 OK; (9+8)(9-8)=17*1=17 OK; (6+4)(6-4)=10*2=20. Answer (a) 20.

Q10. Answer: (b)

A & B: A is daughter of B. B $ C: B is wife of C, so C is A's father. C @ D: C is brother of D, so D is A's uncle/aunt. Hence A is the niece of D. Answer (b) Niece.

Q11. Answer: (c)

Resolve vectors (E=+x). E:(1,0); NE:(1/sqrt2,1/sqrt2); N:(0,1); NW:(-1/sqrt2,1/sqrt2); W:(-1,0). Sum x = 1 -1 + (1/sqrt2 - 1/sqrt2) = 0. Sum y = 1 + 2/sqrt2 = 1 + sqrt2. Displacement = (1+sqrt2) km. Answer (c).

Q12. Answer: (b)

Rule linking 12 and 144 via +1: 12 -> 12*(12) = 144 i.e. n*n. But hidden +1 step => n*(n+1)? 12*? The intended rule n^2 gives 17^2=289. With +1 step n*(n+1): 17*18 = 306. The phrasing 'hidden +1 step' selects 306. Answer (b) 306.

Q13. Answer: (a)

Common face 3 appears in both positions; rotating, faces adjacent to 2 are 1,3,4,6, so opposite 2 is the remaining face 5.

Answer (a) 5.

Q14. Answer: (c)

Square + both diagonals + both midlines creates many triangles; systematic count gives 24. Answer (c).

Q15. Answer: (a)

Col1: B,C,D,E -> F (+1). Col2: D,F,H,J -> L (+2). Col3: F,I,L,O -> R (+3). Next group = FLR. Answer (a) FLR.

Q16. Answer: (a)

A is son of B; B is daughter of C; C is wife of D; D's only son is E. So E is the son of C and D, hence E is B's brother. A is B's son => A is E's nephew. Answer (a).

Q17. Answer: (a)

Pattern: each letter +1 (M->N, A->B, D->E, R->S, A->B, S->T = NBESBT). Apply to BOMBAY: B->C, O->P, M->N, B->C, A->B, Y->Z = CPNCBZ. Answer (a).

Q18. Answer: (a)

Mirror time + actual time = 12:00 (or 11:60). Actual = 11:60 - 2:30 = 9:30. Answer (a).

Q19. Answer: (b)

1331=11^3, 2197=13^3, 4913=17^3, 6859=19^3 are perfect cubes; 4096 = 16^3? 16^3=4096 IS a cube but also 2^12 and a perfect square (64^2). The others are cubes of ODD primes; 4096 is even / 16^3. Odd one = 4096. Answer (b).

Q20. Answer: (a)

C>A>B, B>D>E. So order: C>A>B>D>E. Given C not highest is contradictory unless someone above C; but only these five, so lowest is clearly E. Answer (a) E.

Q21. Answer: (d)

Odd positions (1st,3rd,5th,7th): 1,27,125,343 = 1^3,3^3,5^3,7^3. Even positions (2nd,4th,6th): 4,16,36 = 2^2,4^2,6^2. Both interleaved patterns are correct -> no wrong term. Answer (d).

Q22. Answer: (b)

Ranked (passed & present) = 18 + 28 - 1 = 45. Failed (5) and absent (3) are enrolled but not in this list. Total enrolled = 45 + 5 + 3 = 53. Answer (b) 53.

Q23. Answer: (b)

Square both sides: x+16 = (x-4)^2 = x^2 -8x +16 -> x+16 = x^2 -8x+16 -> x^2 -9x = 0 -> x=0 or 9. Check x-4>=0 (RHS must be >=0): x=9 gives sqrt(25)=5=9-4 valid; x=0 gives sqrt16=4 != -4. Answer (b) 9.

Q24. Answer: (b)

Tree is 5th from each end -> total = 5 + 5 - 1 = 9. Answer (b).

Q25. Answer: (a)

Let each got x. Total after firing: 3x - 12 = x -> 2x = 12 -> x = 6. Original total = 3x = 18. Answer (a).

SECTION B - General Awareness

Q26. Answer: (b)

Certiorari quashes the order of a lower court/tribunal that acted beyond jurisdiction. (Prohibition only prevents; certiorari quashes.)

Q27. Answer: (b)

The Permanent Settlement (1793) was introduced by Lord Cornwallis.

Q28. Answer: (c)

The Tapti (and Narmada) flow through rift valleys between block mountains.

Q29. Answer: (b)

Judicial Review is borrowed from the USA.

Q30. Answer: (c)

Argon is an inert noble gas, not a greenhouse gas.

Q31. Answer: (a)

Quit India Movement was launched on 8 Aug 1942 at the Bombay (Gowalia Tank) session.

Q32. Answer: (b)

Iodine deficiency causes goitre.

Q33. Answer: (c)

Gujarat has the longest coastline among Indian states (~1600 km).

Q34. Answer: (c)

Article 32 - Right to Constitutional Remedies (Ambedkar's 'heart and soul' of the Constitution).

Q35. Answer: (b)

Niels Bohr proposed fixed, non-radiating stationary orbits.

Q36. Answer: (b)

The Tashkent Declaration (1966) was mediated by the USSR (PM Kosygin).

Q37. Answer: (b)

The ozone layer lies in the stratosphere.

Q38. Answer: (b)

Sarojini Naidu was the first Indian woman to preside over the INC (1925). (Annie Besant, 1917, was the first woman but British/Irish-origin.)

Q39. Answer: (c)

Bauxite is the chief ore of aluminium.

Q40. Answer: (b)

Money Bill is defined under Article 110.

Q41. Answer: (b)

The Second Five-Year Plan was based on the Mahalanobis (Gadgil-Mahalanobis) model.

Q42. Answer: (b)

The SI unit of magnetic flux is the weber (Wb).

Q43. Answer: (b)

The Vijayanagara Empire was founded (1336) by Harihara and Bukka (Sangama dynasty).

Q44. Answer: (b)

The Liquidity Adjustment Facility is operated by the RBI (repo/reverse-repo).

Q45. Answer: (b)

The cerebellum regulates balance, posture and coordination.

Q46. Answer: (c)

The Treaty of Srirangapatna (1792) ended the Third Anglo-Mysore War.

Q47. Answer: (b)

Black soil (regur) forms from weathered basaltic lava and is iron-rich.

Q48. Answer: (c)

Aqua regia = conc. HCl + conc. HNO_3 in 3:1 ratio.

Q49. Answer: (b)

Sher Shah Suri built/rebuilt the Grand Trunk Road (Sadak-e-Azam).

Q50. Answer: (b)

El Nino = abnormal warming of central and eastern equatorial Pacific surface waters.

SECTION C - Quantitative Aptitude

Q51. Answer: (c)

(a+b+c)^2 = 36 = 14 + 2(ab+bc+ca) -> ab+bc+ca = 11. a^3+b^3+c^3-3abc = (a+b+c)(a^2+b^2+c^2 - ab-bc-ca) = 6 x (14-11) = 18. Answer (c) 18.

Q52. Answer: (b)

CI: amount(3yr)/amount(2yr) = 2662/2420 = 1.1 = (1+r/100). So r = 10%.

Q53. Answer: (a)

Complementary angles from distances a and b: height h = sqrt(a*b).

Q54. Answer: (b)

Square: (sin+cos)^2 = 2 -> 1 + 2 sin cos = 2 -> sin cos = 1/2.

Q55. Answer: (a)

Together rate 1/12. In 8 days work done = 8/12 = 2/3. Remaining 1/3 done by B in 8 days -> B's rate = (1/3)/8 = 1/24 -> B alone 24 days. A's rate = 1/12 - 1/24 = 1/24 -> A alone 24 days. Answer (a) 24.

Q56. Answer: (a)

Selling two items at same price with +x% and -x%: always a loss of x^2/100 = 100/100 = 1%. Answer (a) loss of 1%.

Q57. Answer: (a)

Old sum = 5000. Corrected sum = 5000 - 9 + 90 = 5081. Mean = 50.81.

Q58. Answer: (a)

LCM(12,16,18,21): 12=2^2*3,16=2^4,18=2*3^2,21=3*7 -> LCM=2^4*3^2*7=1008. Number = 1008 + 7 = 1015.

Q59. Answer: (a)

Net rate = 1/12 + 1/15 - 1/20. LCM 60: (5 + 4 - 3)/60 = 6/60 = 1/10 per min. Time = 10 min. Answer (a) 10.

Q60. Answer: (a)

7^4 = 2401 -> last two digits 01. 2008 = 4*502, so 7^2008 = (7^4)^502 -> last two digits 01. Answer (a) 01.

Q61. Answer: (a)

Length L = 8v (pole). Platform: L+264 = 20v. -> 264 = 12v -> v = 22 m/s = 79.2 km/h. Answer (a).

Q62. Answer: (a)

x = 7-4sqrt3 = (2-sqrt3)^2, so sqrt(x) = 2-sqrt3. 1/sqrt(x) = 1/(2-sqrt3) = 2+sqrt3. Sum = 4. Answer (a) 4.

Q63. Answer: (a)

Horse = h, cart = 3000-h. Net gain 2% of 3000 = 60. 0.20h - 0.10(3000-h) = 60 -> 0.3h - 300 = 60 -> 0.3h = 360 -> h = 1200. Answer (a) Rs.1200.

Q64. Answer: (a)

ALLAHABAD: 9 letters; A appears 4 times, L 2 times, others H,B,D single. = 9!/(4!*2!) = 362880/(24*2) = 362880/48 = 7560. Answer (a).

Q65. Answer: (a)

Roots 2k,3k: sum=5k=p, product=6k^2=q. k=p/5 -> q=6p^2/25 -> 25q = 6p^2. Answer (a) 6p^2 = 25q.

Q66. Answer: (b)

For equal perimeter, the circle encloses the larger area (circle maximises area for given perimeter). Answer (b).

Q67. Answer: (b)

Net = 1 - 0.8*0.75 = 1 - 0.6 = 0.4 -> 40% single discount. Answer (b).

Q68. Answer: (a)

Let 2^x=3^y=6^(-z)=k. x=log k/log2, etc. 1/x=log2/logk, 1/y=log3/logk, 1/z = -log6/logk. Sum = (log2+log3-log6)/logk = log(6/6)/logk = 0. Answer (a) 0.

Q69. Answer: (c)

Volume ratio = (6/2)^3 = 27. Number of small spheres = 27. Answer (c).

Q70. Answer: (a)

At marriage avg 23 => sum 46. After 5 yrs each +5 => sum 56 (two people). Child age 1 => family sum = 56 + 1 = 57, members = 3. Average = 57/3 = 19. Answer (a) 19.

Q71. Answer: (a)

Standard identity: cos^2 A = (m^2 - 1)/(n^2 - 1). Answer (a).

Q72. Answer: (b)

Cross-section area = pi r^2 = (22/7)(3.5)^2 = 38.5 cm^2. Length/min = 5 m/s*60 = 300 m = 30000 cm. Volume = 38.5*30000 = 1,155,000 cm^3 = 1155 litres. Answer (b) 1155.

Q73. Answer: (a)

Growth factor = 1597.20/1452 = 1.1 -> r = 10%. A2 = P(1.1)^2 = 1.21P = 1452 -> P = 1452/1.21 = 1200. Answer (a) Rs.1200.

Q74. Answer: (a)

P(both kings) = C(4,2)/C(52,2) = 6/1326 = 1/221. Answer (a).

Q75. Answer: (a)

Correct hands coincide every 65 5/11 min. This clock does it every 64 min (faster) -> hands meet sooner than they should -> the clock is GAINING time. Answer (a).

SECTION D - English Comprehension

Q76. Answer: (b)

PELLUCID = transparently clear; opposite = MURKY.

Q77. Answer: (b)

PERFUNCTORY = done routinely/superficially = CURSORY.

Q78. Answer: (a)

Correct spelling: BUREAUCRACY.

Q79. Answer: (b)

'Hardly... when' is the correct correlative, not 'Hardly... than'. Error in (b) 'than the train'.

Q80. Answer: (b)

Fear of confined spaces = CLAUSTROPHOBIA.

Q81. Answer: (a)

Correct: 'prevailed ON/UPON him' (to persuade).

Q82. Answer: (b)

TACITURN = uncommunicative; antonym = LOQUACIOUS (talkative).

Q83. Answer: (a)

Exclamatory of sorrow -> 'He exclaimed with sorrow that he had failed.'

Q84. Answer: (b)

'To bell the cat' = to take on a dangerous/difficult task oneself.

Q85. Answer: (b)

CAUSTIC = bitingly sarcastic; fits 'deeply offended.'

Q86. Answer: (a)

Correct: comma before direct speech, full sentence in double quotes: The teacher said, "Stop talking and listen carefully."

Q87. Answer: (b)

Government by the wealthy = PLUTOCRACY.

Q88. Answer: (c)

'The number of' takes a singular verb -> 'has increased', not 'have increased'. Error in (c).

Q89. Answer: (b)

EBULLIENT = cheerful/energetic = EXUBERANT.

Q90. Answer: (b)

Correct: 'acquitted OF the charge.'

Q91. Answer: (a)

Correct spelling: LIAISON.

Q92. Answer: (b)

MAGNANIMOUS = generous/noble; opposite = PETTY.

Q93. Answer: (a)

Words on a tomb = EPITAPH.

Q94. Answer: (b)

QUIXOTIC = idealistic and impractical.

Q95. Answer: (b)

'A poverty of genuine productivity' = absence of real, meaningful output.

Q96. Answer: (b)

Author: 'the contemporary badge of honour is the overflowing calendar' -> a full calendar now signals status.

Q97. Answer: (b)

Field left fallow yields more later -> rest/idleness restores creative capacity.

Q98. Answer: (b)

'Valorisation' = glorification/the act of ascribing high value.

Q99. Answer: (b)

The author is critical and sceptical of the cult of busyness.

Q100. Answer: (c)

Author would agree: deliberate idleness can enhance original thinking.